

Robustness of Graph Construction in Context-Aware IoT Anomaly Detection

Comparative Analysis Report

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Abstract

The original paper, *Context-Aware Anomaly Detection by Community Detection in the Internet of Things*, proposes a high-performing framework based on dynamic multi-edge graph construction, community detection, Heterogeneous Graph Neural Networks (HeteroGNN), and edge-level anomaly detection. This report details our project, which focuses on a critical component of that pipeline: **graph construction**. Rather than modifying the HeteroGNN architecture itself, we investigate whether the final performance depends strongly on how IoT traffic is represented as a graph. We perform an ablation study on the edge types (Network, Context, and Knowledge) to evaluate their distinct contributions to anomaly detection performance.

1 Motivation and Project Idea

The original paper reports very high anomaly detection performance, but the framework strongly depends on how raw network traffic is transformed into a multi-relational graph. This dependency makes graph construction a key methodological step for robustness and generalization across different IoT environments.

Our research question is: *How much does each edge type contribute to anomaly detection performance?* To answer this, we systematically isolate the edge types into four configurations:

- **Configuration A:** Network edges only
- **Configuration B:** Network + Context edges
- **Configuration C:** Network + Knowledge edges
- **Configuration D:** Network + Context + Knowledge edges

By evaluating these configurations, we isolate the impact of different graph representations on Accuracy, Precision, Recall, and F1-score.

2 Preprocessing Comparison

The transition from raw tabular network logs to structured graph data requires strict preprocessing. Upon analyzing both the original paper’s methodology and our custom implementation (exemplified by **Configuration A**), we identified several critical divergences in temporal handling, granularity, and feature inclusion.

2.1 Original Paper Preprocessing (Micro-Temporal Dynamics)

The original framework is designed to closely simulate a real-time IoT environment:

- **Global Chronological Ordering:** The entire dataset is ordered strictly by time before any splitting.
- **Second-by-Second Snapshots:** To capture highly granular network dynamics, the paper utilizes continuous second-by-second time-windows (G_t). This results in an extremely sparse but rapidly evolving dynamic graph sequence.
- **Full Multi-Relational Edges:** Every graph snapshot is constructed using all available feature sets simultaneously (Network, Context, and Knowledge) to map out heterogeneous interactions.

2.2 Our Preprocessing Approach (Macro-Temporal & Ablation)

In our implementation, we designed the preprocessing pipeline to prioritize statistical stability and feature isolation for our ablation study. As demonstrated in our `CONFIG_A_train_for_3edge.ipynb` workflow, our steps are:

1. **Stratified Subsampling:** Rather than taking a raw chronological cut, we first extract a 20% representative subset using stratified sampling. This ensures that our Train and Test sets have the exact same Benign/Attack ratio, preventing performance skews caused by bursty attacks.
2. **Local Chronological Sorting:** After the data is split, we locally sort the subsets by `Timestamp` to re-establish a temporal sequence.
3. **Hourly Aggregation Slices:** Instead of second-by-second snapshots, we aggregate network traffic into **hourly time-slices** (using `pd.Grouper(freq='h')`). This creates fewer, much denser graphs that represent macro-level community behaviors rather than micro-second anomalies.
4. **Feature Isolation (Ablation):** For Configuration A specifically, we strictly filter the raw data to generate *only* Network edges (Src Port, Dst Port, Protocol). Context and Knowledge features are explicitly discarded during edge generation.
5. **Community Detection:** We apply the Label Propagation Algorithm (LPA) to each static hourly slice to generate community labels, which are appended to the nodes as features (`x`) before converting the structure into a PyTorch Geometric `HeteroData` object.

Implications: Our hourly aggregation strategy sacrifices micro-second tracking in favor of dense, structurally rich snapshots. Combined with feature isolation (e.g., Network Only), our pipeline allows us to definitively measure the structural value of specific edge types without the noise of extreme temporal sparsity.

3 GAE Model Structure Comparison

Both works utilize a Graph Autoencoder (GAE) style architecture leveraging Heterogeneous Graph Neural Networks for edge-level anomaly detection.

3.1 Original Framework Structure

The original architecture is a comprehensive pipeline:

- **Community Detection:** Groups nodes into communities to capture macro-level contextual behaviors.
- **HeteroGNN Encoders:** Aggregates information across multiple disparate edge types (Network, Context, Knowledge) simultaneously.
- **Edge-Level Decoder:** Reconstructs edges or classifies existing edges to detect anomalies based on the learned node embeddings.

3.2 Our Model Structure

Our implementation isolates the core graph learning mechanism. We utilize a `HeteroEdgeClassifier` built upon PyTorch Geometric.

- **Architecture:** We use a combination of `HeteroConv` and standard `GCNConv` layers with 64 hidden channels and 32 output channels.
- **Dynamic Convolutions:** Instead of a fixed architecture, our model dynamically builds its convolutional dictionaries based on the `active_edges` configuration passed during initialization (Configs A, B, C, or D).
- **Optimization:** We employ a dynamic optimal thresholding mechanism on the test set to calculate metrics based on the best possible operation point (maximizing the trade-off on the ROC curve).

4 Results Comparison

The original paper demonstrates state-of-the-art results, often pushing F1-scores and Accuracies into the 99% range by leveraging the full multi-relational graph and community context. Our findings present a more nuanced view of *how* those results are achieved.

4.1 Our Empirical Results

Based on our ablation study, we recorded the following metrics for each configuration:

Configuration	AUC	Precision	Recall	F1-score	Accuracy
A: Network Only	0.9752	0.3864	0.4760	0.4265	0.9858
B: Net + Context	0.9773	0.2937	0.5920	0.3926	0.9796
C: Net + Knowledge	0.9781	0.3424	0.5040	0.4078	0.9837
D: Net + Ctx + Kno	0.9776	0.3405	0.5040	0.4065	0.9836

Table 1: Performance metrics across different graph constructions.

4.2 Analysis and Comparison

While the original paper suggests that the combination of all features yields the best anomaly detection framework, our isolated graph construction study reveals critical operational trade-offs:

- **High Confidence vs. High Sensitivity:** Configuration A (Network Only) achieves the highest Precision and F1-score, acting as a highly conservative and accurate alert system. Conversely, adding Context edges (Configuration B) massively boosts Recall (from 47.6% to 59.2%) but severely degrades Precision. Context acts as a *sensitivity multiplier*, catching more hidden attacks but introducing noise.
- **Structural Foundation:** Configuration C (Network + Knowledge) achieves the best underlying mathematical separation (highest AUC at 0.9781) and strikes a balanced middle ground.
- **Redundancy in Full Graphs:** Interestingly, our Configuration D (the equivalent of the original paper’s full graph) does not strictly outperform Configuration C. The performance plateaus, suggesting that simply throwing more relationships (Context + Knowledge simultaneously) at the HeteroGNN creates redundancy rather than pure improvement.

5 Conclusion

Our investigation successfully validates the initial hypothesis: the robustness and performance of context-aware IoT anomaly detection frameworks fundamentally depend on graph construction choices. The original paper’s high performance is not solely a product of the HeteroGNN architecture, but is highly sensitive to the types of edges utilized. By manipulating the graph representation, researchers can fundamentally shift the model’s operational profile from a high-confidence precision filter to a high-sensitivity threat net.